

ORIGINAL RESEARCH

3D medical image segmentation using the serial-parallel convolutional neural network and transformer based on cross-window self-attention

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Abstract

Convolutional neural network (CNN) with the encoder-decoder structure is popular in medical image segmentation due to its excellent local feature extraction ability but it faces limitations in capturing the global feature. The transformer can extract the global information well but adapting it to small medical datasets is challenging and its computational complexity can be heavy. In this work, a serial and parallel network is proposed for the accurate 3D medical image segmentation by combining CNN and transformer and promoting feature interactions across various semantic levels. The core components of the proposed method include the cross window self-attention based transformer (CWST) and multi-scale local enhanced (MLE) modules. The CWST module enhances the global context understanding by partitioning 3D images into non-overlapping windows and calculating sparse global attention between windows. The MLE module selectively fuses features by computing the voxel attention between different branch features, and uses convolution to strengthen the dense local information. The experiments on the prostate, atrium, and pancreas MR/CT image datasets consistently demonstrate the advantage of the proposed method over six popular segmentation models in both qualitative evaluation and quantitative indexes such as dice similarity coefficient, Intersection over Union, 95% Hausdorff distance and average symmetric surface distance.

KEYWORDS

convolution neural network, cross window self-attention, medical image segmentation, transformer

1 | INTRODUCTION

Medical image segmentation involves extracting the tissues, organs, and pathological regions from the complex backgrounds of medical images [1]. Leveraging the cutting-edge computer-assisted techniques, medical image segmentation

can assist in disease diagnosis, lesion localisation, treatment planning and surgical navigation. However, due to the complexity, variability and noise of the medical images, it is a great challenge to achieve accurate and robust segmentation.

Over the past decade, deep learning has been explored for image segmentation because of its excellent ability to handle

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the complex image data. Unlike the traditional methods which require the manual feature design [2–4], deep learning enables the end-to-end learning from data. U-Net [5] currently dominates medical image segmentation by integrating various semantic features based on the symmetrical encoder–decoder architecture. As convolutional neural network (CNN) flourishes in computer vision, innovations from visual contexts have been seamlessly incorporated to refine medical image segmentation models [6–9]. However, the CNN is constrained by the limited local receptive fields despite the incorporation of techniques such as feature pyramid networks [10] and dilated convolutions [11, 12] that enhance network capabilities. In clinical practice, the shapes of organs and tumours may be influenced by surrounding tissues, underscoring the value of extracting the global information for improved segmentation accuracy.

Recently, the transformer [13] has gained attention in computer vision tasks because it can extract long-distance dependency from sequence data. By employing self-attention mechanisms, the transformer dynamically assigns weights to sequence elements, resulting in a broad receptive field and enhanced contextual information capture. The pure transformer model has been explored for the medical segmentation task. However, the application of transformer to 3D medical images requires careful consideration of challenges posed by spatial and computational complexity.

Because the CNN and transformer exhibit different feature extraction capabilities in image segmentation tasks, it is significant to combine CNN and transformer in the encoder–decoder structure [14, 15]. Along this line, this paper has proposed the serial and parallel CNN and transformer (SPCT) network, a hybrid approach that combines convolution and self-attention for volumetric segmentation. The SPCT consists of a serial CNN encoder, parallel transformer blocks, and a cascaded convolution decoder. The summarised contributions are as follows:

- 1) We have proposed a novel architecture that effectively synergises the strengths of CNN and transformer, enhancing feature interactions across diverse semantic levels.
- 2) We have introduced the cross-window self-attention block, which partitions 3D images into non-overlapping windows and computes sparse global attention across these windows. This innovation significantly improves the model's ability to capture the global information under the affordable computational burden.
- 3) We have introduced the multi-scale local enhanced (MLE) block, which efficiently computes attention between various parallel branches to fuse different semantic features and utilises convolution to enhance the local feature extraction.
- 4) We have done extensive experiments on prostate, atrium and pancreas datasets to prove the advantage of our model over other segmentation models in segmenting various organs in MR/CT images.

2 | RELATED WORK

2.1 | Convolutional neural network for medical image segmentation

CNN plays an important role in medical image segmentation. Among the CNN models, the U-Net [5] architecture characterised by its U-shaped design and skip connections serves as a notable exemplar. Here, the skip connections fuse different information between the decoder and encoder, enabling the effective delineation of anatomical structures. Furthermore, attention U-Net [16] introduces dynamic attention gates to enhance detail capture, particularly in complex scenarios such as lesion identification. Res-UNet [17] employs residual connections to address the vanishing gradient challenge in deep networks, improving the gradient flow during training. U-Net++ [18] extends the U-Net paradigm by incorporating nested pathways for capturing features at various scales, which proves to be helpful for segmenting the intricate anatomical structures. UNet3+ [19] utilises full-scale skip connections to retain high-resolution features from different levels and improve segmentation accuracy. For automated segmentation, nnUNet [20] streamlines the process and reduces the need for manual intervention through an end-to-end approach. Its comprehensive architecture covers various preprocessing steps, network configurations, and post-processing strategies, making it suitable for various segmentation tasks.

2.2 | Transformer models for image segmentation

Transformer models have led to a significant paradigm shift in computer vision. The vision transformer [21], pioneering the integration of transformer architecture for image classification, not only expands its horizons beyond natural language processing but also serves as a catalyst for further exploration in computer vision. Notably, the Swin transformer [22] introduces a hierarchical framework with stages and window attention mechanism, effectively capturing both global and local contextual information. This innovation makes the Swin transformer a powerful backbone for diverse visual tasks such as image classification, object detection, and image segmentation.

In semantic segmentation, the transformer has the ability of preserving consistent spatial resolution between inputs and outputs. When images are represented as semantic tokens, the transformer can model the relationships between tokens and subsequently decode predictions. Consequently, models such as Swin-Unet [23], SegFormer [24], Segmenter [25], and SETR [26] are proposed, and they demonstrate the outstanding performance in segmentation tasks. The Swin-Unet adopts Swin transformer's modules as its fundamental units. With a symmetric structure that includes upsampling operations and patch expansion layers, the Swin-Unet achieves both local-to-global attention modelling and feature map restoration. The SegFormer integrates the transformer with lightweight

multilayer perceptron (MLP) decoders. It employs a hierarchically structured transformer encoder that generates multi-scale features without requiring positional embeddings. The Segmenter utilises class tokens and patch embeddings for self-attention, and subsequently multiplies class tokens with output embeddings to make predictions. The SETR revolutionises semantic segmentation by treating the image as a sequence of patches and implementing the pure transformer architecture for image encoding, thereby reshaping the segmentation process into the sequence-to-sequence task.

2.3 | CNN-transformer hybrid models for medical image segmentation

Compared with CNNs, transformers have powerful global feature representation ability, but need more data for training. The limited medical dataset size poses a challenge for transformers. To address this limitation, researchers have been actively exploring hybrid models that combine the strengths of both convolution and self-attention. These hybrid models use CNNs to learn the local image features and transformers to learn the global contextual information. For instance, TransBTS [27] integrates transformers into the U-Net bottleneck, and applies self-attention effectively by reducing sequence length. Similarly, TransUNet's [28] encoder combines shallow CNN layers and deep transformers, where image blocks are converted into tokens via CNNs. Besides, UNETR [29] captures the global multi-scale information effectively by using the transformer as the encoder, and establishes direct connections between different transformer layers and the CNN decoder for accurate segmentation. UNETR++ [30] extends UNETR with efficient paired attention blocks, and merges spatial and channel attention for high-quality segmentation. To optimise computation, CoTr [31] introduces a deformable transformer module to capture long-range dependencies effectively. Unlike the traditional transformer, CoTr only focuses the deformable self-attention on the key points, thereby simplifying the processing of high-resolution feature maps. Derived from nnUnet, nnFormer [32] is a semantic segmentation framework that combines self-attention and convolution. It employs skip attention for feature fusion and utilises interleaved transformer and convolution down-sampling blocks. This boosts model adaptability, and enhances the generalisation and performance.

The above innovations demonstrate the advancement of CNN-transformer networks and the continuous exploration of hybrid models for improving medical image segmentation accuracy.

3 | METHOD

3.1 | Network structure

The network structure of our SPCT model is shown in Figure 1. The network includes three core components: the CNN encoder, transformer with feature interaction modules,

and lightweight segmentation decoder. In detail, the encoder is a modified 3D ResNeXt [33], where we have adjusted the initial convolution layer's stride from 2 to 1 to reduce the down-sampling rate. The convolutional blocks of the ResNeXt consist of Conv–BN–ReLU units with residual connections in a bottleneck shape. These blocks incorporate group convolutions to learn diverse features from the distinct low-dimensional subspaces. From the last three convolutional layers, the features will be produced with down-sampling ratios of {4, 8, 16}. Subsequently, our architecture incorporates the FPN [10] structure to fuse features from various resolutions through top–down integration. Upsampling is applied to merge features of different resolutions to produce the fused features.

After the FPN structure, the designed cross window self-attention (CWSA) based transformer (CWST) block (discussed in section 3.2) is used to learn long-range dependencies. Notably, the transformers for different branches have separate parameters and capture information specific to their respective feature levels. After the first transformer block, we employ the MLE blocks (discussed in section 3.3) to interactively fuse the features from the three branches. This process combines both serial and parallel operations to sequentially extract features from individual levels and then interact with multi-level features.

Finally, a simple segmentation decoder is employed, which involves the cascaded CNN block to concatenate features from different branches along the channel dimension. The final prediction is achieved through convolution and upsampling. To accelerate convergence and introduce regularisation, deep supervision is applied to the outputs of the three branches.

3.2 | The cross window self-attention based transformer block

In the CNN encoder, the limited receptive fields of convolutional kernels render it difficult to capture the global information. Although the FPN can be used to fuse multi-scale features, it may not be enough to obtain the desirable global context. To address this issue, we introduce the transformer. Considering that performing attention computations on each voxel of a 3D image directly can be computationally burdensome, we have proposed a novel window attention method, which learns the sparse global information from down-sampled sub-images to establish the overall global context for the task.

Dividing and flattening: After selecting the appropriate window size, our initial step involves dividing the input $\in \mathbb{R}^{C \times H \times W \times D}$ into non-overlapping windows, where each window is a cube of size $ph \times pw \times pd$. The number of windows is $N = (H \times W \times D)/(ph \times pw \times pd)$. As shown at the bottom of Figure 1, elements that occupy matching positions (indicated by the same colour) in different windows are then flattened into the individual sequences, yielding $S = ph \times pw \times pd$ sequences. This transformation produces a flattened image of size $S \times N \times C$, thereby creating a perspective in which each sequence corresponds to a sub-image.

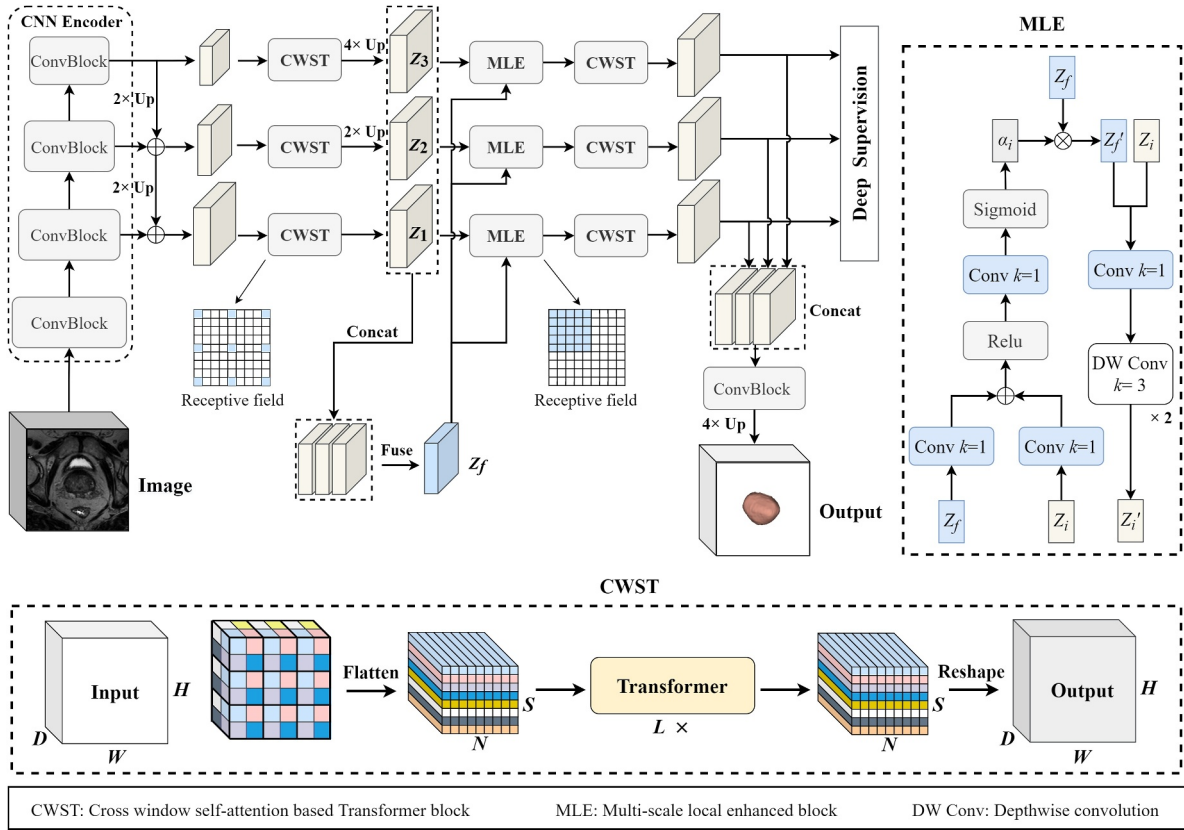


FIGURE 1 Overview of our SPCT network. The initial CNN encoder on the left transforms the 3D image into multi-scale features, which are then fed to the FPN for preliminary fusion. Subsequently, the first-level CWST blocks are used for capturing the global information to produce $\{Z_1, Z_2, Z_3\}$. After the first-level transformer blocks, the features are upsampled, concatenated and fused into a guiding feature Z_f . Z_f and Z_i are input to the MLE blocks. The outputs from the second-level CWST blocks are fed into the segmentation head, a cascaded CNN, to generate predictions through upsampling, CNN, convolutional neural network; CWSA, cross window self-attention; FPN, feature pyramid network; MLE, multi-scale local enhanced; SPCT, serial and parallel CNN and transformer.

Linear embedding: The flattened tensor is projected into *Query*, *Key*, and *Value* representations using three linear layers W_Q , W_K and W_V . Since our input consists of multiple sequences, the output of these linear layers introduces an extra dimension compared with the traditional transformers [13]. We reshape the input to create a sequence for each set of voxels within a window. In other words, we have S sequences (sub-images). The linear projection is defined as follows:

$$\begin{aligned} Q_i &= W_Q \cdot \text{LN}(x_i) \in \mathbb{R}^{S \times N \times d_k} \\ K_i &= W_K \cdot \text{LN}(x_i) \in \mathbb{R}^{S \times N \times d_k} \\ V_i &= W_V \cdot \text{LN}(x_i) \in \mathbb{R}^{S \times N \times d_k} \end{aligned} \quad (1)$$

where Q , K , V represent the weight matrices for *Query*, *Key*, and *Value*, respectively; x denotes the input feature, and LN denotes layer normalisation for standardising the features at each position for stable training.

Cross window self-attention: Originally, multi-head self-attention [13] aims to project features into low-dimensional embeddings to gather diverse information. In our approach, we spatially decompose high-resolution features into multiple low-resolution ones to facilitate extracting the global information more efficiently. Each element in the low-resolution

features for self-attention computation comes from different windows, so the CWSA can be calculated. This concept shares similarity with the dilated convolutions, where the convolutional kernel is expanded to increase the receptive field. However, the dilated convolutions require manual setting of dilation rates and they are troubled by the gridding effect. In contrast, our CWSA block guarantees a global receptive field for each sequence. As depicted in Figure 1, we compute attention for the i th voxel in a window and that in the remaining $N-1$ windows, and repeat this process for S times. The computation for CWSA is as follows:

$$\text{CWSA}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

where d_k means the dimension of the projected *Key*, and Softmax is applied to the last dimension of the attention matrix $\text{attn} \in \mathbb{R}^{S \times N \times N}$.

Transformer layers: We employ the transformer blocks in the three branches, where each works independently with its own set of parameters. These branches handle image features of various resolutions. To reduce computation, we use different window sizes and prioritise larger ones for higher-resolution

features. Naturally, the specific datasets may exhibit varying image dimensions. We can customise down-sampling ratios, branch quantities, and block configurations accordingly. In Figure 1, the window sizes for the first three transformer blocks are set as $4 \times 4 \times 4$, $2 \times 2 \times 2$, and $1 \times 1 \times 1$, respectively. Notably, a $1 \times 1 \times 1$ window size corresponds to standard self-attention [21]. The outputs of the initial transformer block are denoted as $\{Z_1, Z_2, Z_3\}$. We upsample these features to a uniform size for interaction, and then feed them into the second transformer block. The subsequent three blocks consistently maintain a window size of $4 \times 4 \times 4$. The transformer layer includes CWSA and MLP modules, whose outputs can be given using the following equation:

$$\begin{aligned} \hat{z}_l &= \text{CWSA}(\text{LN}(z^{l-1}) + z^{l-1}) \\ z_l &= \text{MLP}(\text{LN}(\hat{z}_l)) + \hat{z}_l \end{aligned} \quad (3)$$

where $\hat{z}_l$ and z^l denote the outputs from the CWSA and MLP at layer l , respectively.

Different self-attention mechanism: In Figure 2, we present four attention modules for 3D images: (a) spatial self-attention (SA) [21], (b) axial self-attention (ASA) [34], (c) WSA [22] and (d) CWSA. Among these modules, the SA treats each voxel as a token, and captures dependencies across all voxels. However, its computational cost is expensive, especially for high-resolution images. The ASA takes a different strategy by computing self-attention iteratively along each dimension, namely height, width, and depth. Although this approach improves computational efficiency, it may sacrifice some modelling capacity as it does not consider interactions between voxels outside the current dimension.

The WSA and our CWSA are proposed to reduce the computation cost while maintaining acceptable performance degradation. The two modules employ image partitioning techniques to reduce sequence length, which is reduced by the WSA and the CWSA to S (window area/volume) and N (number of windows), respectively. From a sub-image perspective, the former calculates self-attention for the localised images ($ph \times pw \times pd$) while the latter targets the down-sampled images ($H/ph \times W/pw \times D/pd$). The receptive field of the WSA is the same as the window size and that of the CWSA is global but sparse. Consequently, Swin transformer

[22] learns inter-window information through window shifting, whereas our method leverages convolution (discussed in section 3.3) to capture the densely localised details. The computational complexities Ω of the four attention modules are given using the following equation:

$$\begin{aligned} \Omega(\text{SA}) &= (H \cdot W \cdot D)^2 C \\ \Omega(\text{ASA}) &= (H \cdot W \cdot D)(H \cdot W + H \cdot D + W \cdot D)C \\ \Omega(\text{WSA}) &= (H \cdot W \cdot D)^2 C/N \\ \Omega(\text{CWSA}) &= (H \cdot W \cdot D)^2 C/S \end{aligned} \quad (4)$$

where the spatial dimensions of the volume are represented as (H, W, D) and the channel dimension is denoted as C . Accordingly, the WSA and CWSA reduce the computational load to $1/N$ and $1/S$ of the SA, respectively.

3.3 | Multi-scale local enhanced block

After the first level transformer blocks, the features from the three branches are output as $\{Z_1, Z_2, Z_3\}$. These features are concatenated along the channel dimension and then are sent to sequential $1 \times 1 \times 1$ and $3 \times 3 \times 3$ convolutions to obtain the fused feature Z_f . Here, Z_f plays a role in guiding and reinforcing the features within the three branches. Subsequently, Z_f and Z_i are input into the MLE block as illustrated in upper right of Figure 1. By means of MLE block, the voxel attention coefficient α_i is computed as follows:

$$\alpha_i = \text{Sigmoid}(W_m(\text{Relu}(W_i Z_i + W_f Z_f))) \quad (5)$$

where the attention coefficient selectively enhances the feature responses of Z_i with respect to Z_f . W_m is a $1 \times 1 \times 1$ convolution, which reduces the number of channels to 1; W_i and W_f are $1 \times 1 \times 1$ convolutions employed to align Z_i and Z_f in terms of channels.

Next, the attention coefficient α_i is multiplied with Z_f and concatenated with Z_i . This concatenated result is then fed into the convolution block to extract the local features. We employ two layers of $3 \times 3 \times 3$ depth-wise convolution with a $5 \times 5 \times 5$ receptive field, where the maximum window size is set to $4 \times 4 \times 4$. This design ensures that every element within

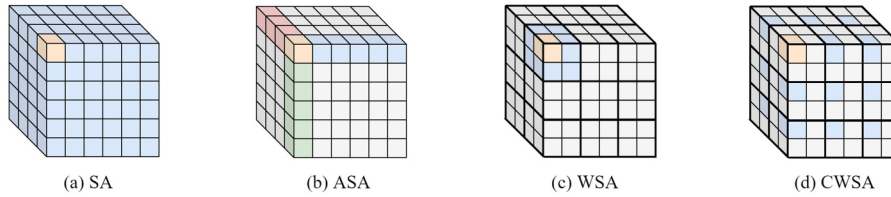


FIGURE 2 Various 3D self-attention modules for volumes. Yellow cubes represent the selected voxels while blue cubes represent the queried regions for SA, WSA and CWSA. Blue, green, and red cubes indicate the queried regions along each axis for ASA. The WSA computes attention within the defined windows while the CWSA calculates attention between different windows. Both WSA and CWSA use a window size of $2 \times 2 \times 2$. ASA, axial self-attention; CWSA, cross window self-attention; WSA, window self-attention.

each voxel establishes dependencies with all elements inside the window so that the limitation of the CWSA in capturing the dense local information can be addressed.

$$Z'_i = \text{DWConv}(\text{Cat}\{Z_i, Z_f * \alpha_i\}) \quad (6)$$

where the element-wise multiplication is performed between Z_f and α_i , Cat denotes concatenation and DWConv represents the depth-wise convolution.

3.4 | Loss function

To ensure the training effectiveness of the proposed network, a hybrid loss function is employed by combining Dice loss [35] and cross-entropy loss. Dice loss is used to measure the degree of overlap between the predictions and ground truths, thereby enabling the model to maintain spatial consistency. On the other hand, the cross-entropy loss assigns a class label to each voxel for voxel-level classification. Its advantage lies in providing clear gradient directions, facilitating rapid convergence and achieving accurate classification results. This hybrid loss function encourages the model to learn the detailed segmentation while ensuring accurate voxel-level classification, and it is computed as follows:

$$L = 1 - \frac{2}{M} \sum_{j=1}^M \frac{\sum_{i=1}^R P_{ij} G_{ij}}{\sum_{i=1}^R P_{ij}^2 + \sum_{i=1}^R G_{ij}^2} - \frac{1}{R} \sum_{j=1}^M \sum_{i=1}^R G_{ij} \log P_{ij} \quad (7)$$

where M and R represent the number of classes and voxels in the input volume, respectively; P_{ij} and G_{ij} denote the model's predictions and ground truths at voxel i for class j , respectively.

The features in the three branches are well-suited for deep supervision due to their reinforcement and mutual interaction. We upsample the branch outputs, compute losses between the outputs and ground truths, and then add these branch-specific losses to the segmentation loss. The deep supervision loss L_{total} is calculated as the summation of losses from both the segmentation head and the three parallel branches:

$$L_{total} = w_{seg} L_{seg} + \sum_{i=1}^3 w_i L_i \quad (8)$$

where L_{seg} is the loss between the prediction and ground truth, and L_i denotes the loss between the output from the i th branch and ground truth. w_{seg} and $w_i = 1, 2, 3$ are the weights of losses from the segmentation head and the three branches, and they are empirically set as (1.0, 0.2, 0.3, 0.4).

4 | EXPERIMENTS

4.1 | Datasets

We have evaluated our method on three public datasets, including the SPIE-AAPM-NCI ProstateX Challenge dataset [36], 2018 Atria Segmentation Challenge dataset [37] as well as pancreas CT dataset [38]. The ProstateX dataset consists of multiparametric magnetic resonance imaging (mpMRI) data and clinical information of patients suspected of prostate cancer. For the mpMRI data, we employ a subset of the dataset containing 914 cases of T2-weighted imaging (T2WI) with a size of $256 \times 256 \times 60$ and a resolution of $0.66 \times 0.66 \times 1.50 \text{ mm}^3$. We randomly divide this subset into 639 cases for training, 137 cases for validation and 138 cases for testing. The left atrium dataset includes 154 gadolinium-enhanced MRI (GE-MRI) scans with a resolution of $0.625 \times 0.625 \times 0.625 \text{ mm}^3$. The scans may vary in size for different patients, but each scan comprises 88 slices. Similarly, the dataset is randomly divided into 100 cases for training, 27 cases for validation and 27 cases for testing. In addition, the pancreas dataset comprises 82 scans. We split the dataset into 56 scans for training, 13 scans for validation and 13 scans for testing. Although the thickness of each scan varies, the resolution of each scan is $0.56 \times 0.56 \times 3.00 \text{ mm}^3$ and the size of each slice is 512×512 pixels.

4.2 | Implementation details

During the model training, the regular online data augmentation techniques have been adopted, including random flipping, rotation, and cropping. All cropping operations are centred around the target organs. Specifically, we perform random cropping on the ProstateX dataset to a size of $192 \times 192 \times 48$. For the left atrium dataset, we implement a crop on it to a size of $128 \times 128 \times 80$. As regards the pancreas CT dataset, we resample all CT images to the same resolution of $1.0 \times 1.0 \times 1.0 \text{ mm}^3$, and their intensities are truncated to the range of $[-125, 275]$ Hounsfield units. During training, the random cropping of $96 \times 96 \times 96$ around the pancreas is implemented. Additionally, z-score normalisation is applied to all three datasets for preprocessing.

The framework is based on the MindSpore Lite tool [39]. In order to accelerate training process, NVIDIA GeForce A100 GPU with 40 GB memory is used. We optimise the network using AdamW, where the weight decay is chosen as 0.01 and the optimiser momentum parameters β_1 and β_2 are set to 0.9 and 0.999, respectively. The learning rate is initialised as 0.001 and adjusted to 0.0001 using cosine decay during training. The same batch size of 4 and the same datasets are used for training, validation, and testing of our method and the compared algorithms. All methods are trained until convergence on the prostate, atrium, and pancreas datasets. Following references [40–42], we employ a sliding inference technique to

segment the entire image during model testing, where a $16 \times 16 \times 4$ stride is used for the prostate and atrium datasets and a $16 \times 16 \times 16$ stride is used for the pancreas dataset.

4.3 | Compared methods and evaluation metrics

Our method is compared with six popular CNN segmentation models and hybrid CNN-transformer models. The former include U-Net [5] and attention U-Net [16], and the latter include TransBTS [27], Swin-UNETR [43], CoTr [31] and nnFormer [32]. To evaluate the performance of all these segmentation methods, we utilise quantitative metrics such as the Dice similarity coefficient (DSC) [35], Intersection over Union (IoU) [44], 95% Hausdorff distance (HD95) [45], and average symmetric surface distance (ASD) [46]. The DSC and IoU evaluate the overlap between predictions and ground truths, which are defined as follows:

$$\text{DSC} = 2 \cdot \frac{|B \cap G|}{|B| + |G|} \quad (9)$$

$$\text{IoU} = \frac{|B \cap G|}{|B \cup G|} \quad (10)$$

where B denotes the binary prediction map and G denotes the ground truth.

Additionally, HD95 quantifies the maximum spatial deviation between segmentations and ASD measures the mean boundary discrepancies between the two segmentations, which are defined as follows:

$$\text{HD}_{95} = \max_{95\%}(b(B, G), b(G, B)) \quad (11)$$

$$\text{ASD} = \frac{1}{R} \sum_{i=1}^R \left(\frac{d_i(B, G) + d_i(G, B)}{2} \right) \quad (12)$$

where $b(B, G)$ represents the 95th percentile distance from the predicted voxels to the ground truth; $d_i(B, G)$ denotes the nearest distance from the predicted voxel to the ground truth; $b(G, B)$ and $d_i(G, B)$ can be defined in a similar way.

4.4 | Comparison on the ProstateX dataset

Table 1 presents the quantitative results of various methods on the ProstateX MRI dataset. U-Net achieves a DSC of 90.91% and attention U-Net performs slightly better than U-Net by providing a DSC of 91.10%. The second-best method is Swin-UNETR, which achieves a DSC of 91.39%. As for our method, it surpasses all other methods by achieving an impressive DSC of 92.23% and an IoU of 85.69%. Notably, our method also demonstrates superior accuracy in boundary delineation with the lowest HD95 (2.76) and the smallest ASD

(1.01) than compared methods. Moreover, our method has the fewest floating-point operations (FLOPs) and parameters among CNN-transformer hybrid methods. Although the segmentation metrics of Swin-UNETR is close to our method, the former has much higher computational load.

Figure 3 presents the 2D and 3D views of the prostate predictions (in red) and ground truths (in green). From the 3D perspective, we can see that the prediction of our method exhibits the highest similarity to the ground truth. In addition, in 2D contour views, the contour of the prediction map by our model closely aligns with that of the ground truth. Indeed, the visual evaluations are consistent with the quantitative results.

4.5 | Comparison on the left atrium dataset

Table 2 and Figure 4 present the quantitative metrics and segmented results on the left atrium dataset, respectively. We can see from Table 2 that our method outperforms the most competitive Swin-UNETR by providing 0.9% higher DSC and 1.44% higher IoU. In Figure 4, other compared methods produce segmentation errors to different extent. By contrast, our method's predictions exhibit better alignment with the ground truth although it is difficult to segment atrial chambers from GE-MRI scans due to its irregular protruding structure and low tissue contrast. Clearly, our method is provided with the lowest computational load, and even surpasses U-Net. This can be attributed to the transformer-based model's faster reduction in FLOPs as the input image size decreases, where FLOPs are associated with the square of the sequence length.

4.6 | Comparison on the pancreas dataset

Table 3 lists the quantitative metrics of all evaluated models on the pancreas CT dataset. Obviously, the U-Net and TransBTS perform worse than other methods in terms of four metrics. Attention U-Net exhibits an improvement over U-Net, increasing the DSC by 1.19%. Moreover, the Swin-UNETR, CoTr, and nnFormer CNN-transformer hybrid models all achieve DSC values over 78%. Our method outperforms the second-best method, Swin-UNETR, with a 1.30% higher DSC, a 1.79% higher IoU and a 0.68 lower HD95, which indeed demonstrates the superiority of our SPCT model.

Visually, Figure 5 shows the segmented results of various methods in 2D and 3D views. Clearly, all methods except the nnFormer and our method produce the isolated regions. Especially, the U-Net even produces the obvious under-segmentation. Although the nnFormer can produce continuous segmented result, our method's prediction has higher degree of overlap with ground truth. The superiority of the SPCT model lies in its outperforming multi-scale information learning ability, which enables it to address the disadvantageous influence of the irregular anatomical structure, the various shapes of pancreas as well as the diverse imaging characteristics such as contrast and resolution.

TABLE 1 Quantitative results for all evaluated models on the ProstateX MRI dataset.

Methods	DSC (%)↑	IoU (%)↑	HD95 (voxels)↓	ASD (voxels)↓	Inference Time (s)↓	FLOPs (G)↓	Params (M)
U-Net	90.91	83.49	3.33	1.25	6.24	584.10	25.89
Attention U-Net	91.10	83.64	3.37	1.23	6.41	594.98	26.15
TransBTS	91.14	83.88	3.07	1.19	8.45	698.65	76.33
Swin-UNETR	91.39	84.30	2.98	1.16	12.81	892.65	62.03
CoTr	91.20	83.97	3.41	1.32	10.72	735.69	41.86
nnFormer	91.28	84.12	3.02	1.17	12.51	854.52	149.23
SPCT	92.23	85.69	2.76	1.01	6.98	647.20	14.84

Note: The bold values indicate the best value for each quantitative metric.
Abbreviations: ASD, average symmetric surface distance; DSC, dice similarity coefficient ; FLOPs, floating-point operations; HD95, Hausdorff distance; IoU, Intersection over Union.

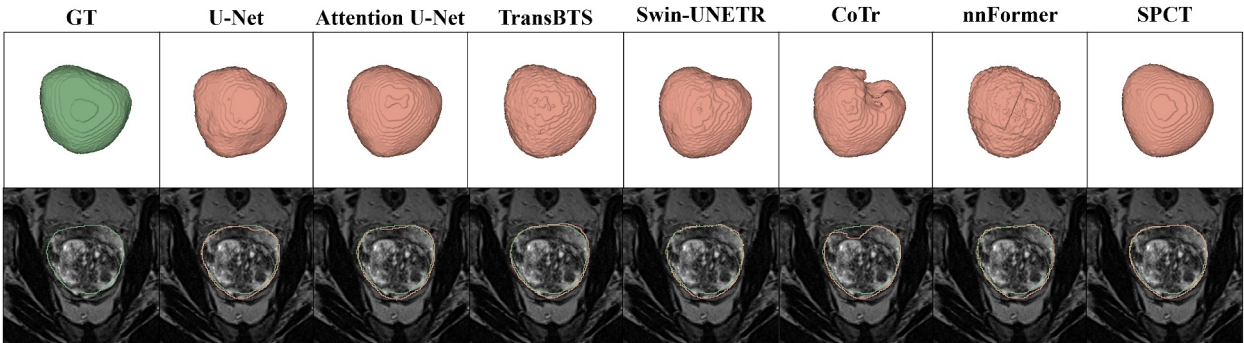


FIGURE 3 3D and 2D views of segmented results from various methods on the ProstateX MRI dataset.

TABLE 2 Quantitative results for all evaluated models on the left atrium MRI dataset.

Methods	DSC (%)↑	IoU (%)↑	HD95 (voxels)↓	ASD (voxels)↓	FLOPs (G)↓
U-Net	90.67	82.94	4.36	1.57	428.67
Attention U-Net	91.45	84.33	5.40	1.46	432.13
TransBTS	91.06	83.59	5.83	1.62	502.88
Swin-UNETR	91.94	85.19	4.66	1.40	534.38
CoTr	91.60	84.50	4.58	1.23	448.29
nnFormer	91.77	84.90	4.82	1.39	532.76
SPCT	92.84	86.63	4.41	1.18	298.52

Note: The bold values indicate the best value for each quantitative metric.
Abbreviation: ASD, average symmetric surface distance; DSC, dice similarity coefficient ; FLOPs, floating-point operations; HD95, Hausdorff distance; and IoU, Intersection over Union.

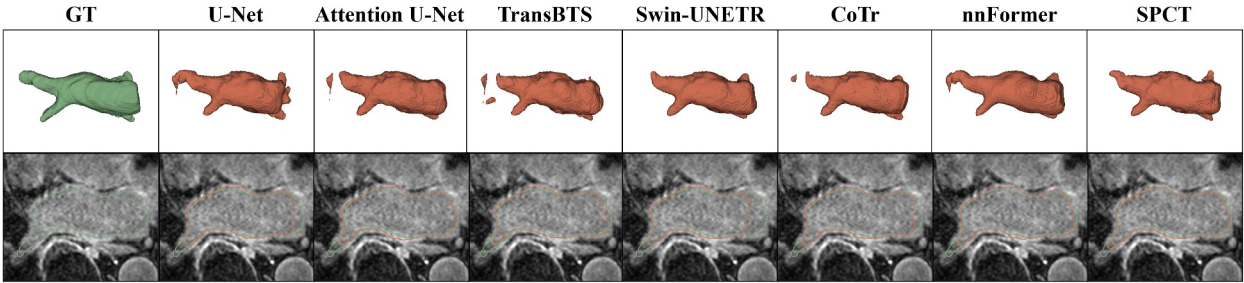


FIGURE 4 3D and 2D views of segmented results from the various methods on the left atrium MRI dataset.

4.7 | Ablation study

We analyse the impact of the CWST and MLE modules in our network on the segmentation of prostate dataset in our ablation experiments. As shown in Table 4, we evaluate our model using four distinct quantitative metrics. Here, the baseline model consists of ResNeXt with FPN and its transformer block is replaced by a single convolution layer. This model demonstrates inferior performance to other four models because it loses some critical details and cannot extract the global context information effectively. For simplicity, we use ASA, WSA, and CWSA to represent transformer blocks with different attention mechanisms.

Compared with the baseline, the introduction of the WSA leads to a 1.82% increase in DSC and a 2.38 reduction in HD95. When the baseline model is only equipped with the CWSA, it outperforms the model only with the WSA by achieving a 0.53% improvement in DSC due to the effective utilisation of sparse global information from the CWSA and dense local information from the CNN encoder. This does not mean that the CWSA has better feature representation ability than the WSA. It might be due to that the CWSA is more suitable for our network structure. As for the ASA, improvements are marginal in both DSC and HD95.

Adding the MLE module to the baseline model improves DSC by 1.42% and reduces HD95 by 1.95. Furthermore, it can improve segmentation accuracy for various transformer configurations. With the introduction of the MLE module, the model with the WSA achieves 0.74% higher DSC and 1.14 lower 95HD while the model with the CWST achieves 1.03%

higher DSC and 0.70 lower 95HD. To further validate our MLE block, we visualise feature maps and attention maps using prostate MR images in Figure 6.

In Figure 6, $\{Z_1, Z_2, Z_3\}$ are features from the three branches after the first-level transformer blocks and are upsampled to a uniform size. Z_f is a fused feature obtained by concatenating $\{Z_1, Z_2, Z_3\}$ along the channel dimension and applying convolution. All features share the same dimensions. Z_f acts as a segmentation prototype with a strong response in the prostate region. $\{\alpha_1, \alpha_2, \alpha_3\}$ are attention maps obtained through voxel-wise attention between Z_f and $\{Z_1, Z_2, Z_3\}$. These maps illustrate how our model progressively focuses its attention from the surrounding tissues to the prostate, captures fine details and refines the prostate's contour. These visualizations confirm our model's effective integration of multi-scale information through parallel branches, thereby improving segmentation accuracy. Figure 7 visually shows the segmented results of the baseline model and its variants on the ProstateX MRI dataset, showcasing the advantage of combining baseline with CWSA and MLE.

Indeed, the above experiments demonstrate the importance and effectiveness of the CWST and MLE modules in improving segmentation accuracy and overall model performance.

4.8 | Discussion

The success of our SPCT model primarily stems from two aspects. Firstly, the CWST module in our model can capture

TABLE 3 Quantitative results for all evaluated models on the pancreas CT dataset.

Methods	DSC (%)↑	IoU (%)↑	HD95 (voxels)↓	ASD (voxels)↓	FLOPs (G)↓
U-Net	77.46	63.21	20.47	2.20	289.35
Attention U-Net	78.65	64.82	14.22	1.49	291.69
TransBTS	76.33	61.73	25.78	1.72	345.69
Swin-UNETR	79.75	66.36	12.34	1.33	359.96
CoTr	78.69	64.87	17.58	1.28	290.78
nnFormer	78.95	65.82	13.84	1.39	329.04
SPCT	81.05	68.15	11.66	1.23	198.01

Note: The bold values indicate the best value for each quantitative metric.

Abbreviations: ASD, average symmetric surface distance; DSC, dice similarity coefficient; FLOPs, floating-point operations; HD95, Hausdorff distance; IoU, Intersection over Union.

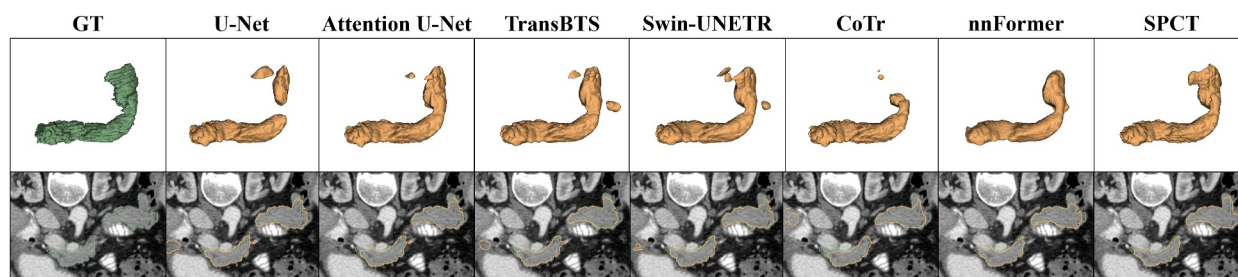


FIGURE 5 3D and 2D views of segmented results from the various methods on the pancreas CT dataset.

TABLE 4 Metrics of baseline and its variants on the ProstateX MRI dataset.

Models	DSC (%)↑	HD95 (voxels)↓	FLOPs (G)↓	Params (M)
ResNeXt + FPN (baseline)	88.85	5.84	296.95	11.68
Baseline + ASA	89.29	4.40	371.18	13.89
Baseline + WSA	90.67	4.24	395.30	13.51
Baseline + CWSA	91.20	3.46	573.75	13.51
Baseline + MLE	90.27	3.89	370.40	13.01
Baseline + ASA + MLE	90.87	3.42	444.63	15.23
Baseline + WSA + MLE	91.41	3.10	468.75	14.84
Baseline + CWSA + MLE	92.23	2.76	647.20	14.84

Abbreviation: ASA, axial self-attention; CWSA, cross window self-attention; DSC, dice similarity coefficient; FLOPs, floating-point operations; FPN, feature pyramid network; HD95, Hausdorff distance; IoU, Intersection over Union; MLE, multi-scale local enhanced; WSA, window self-attention.

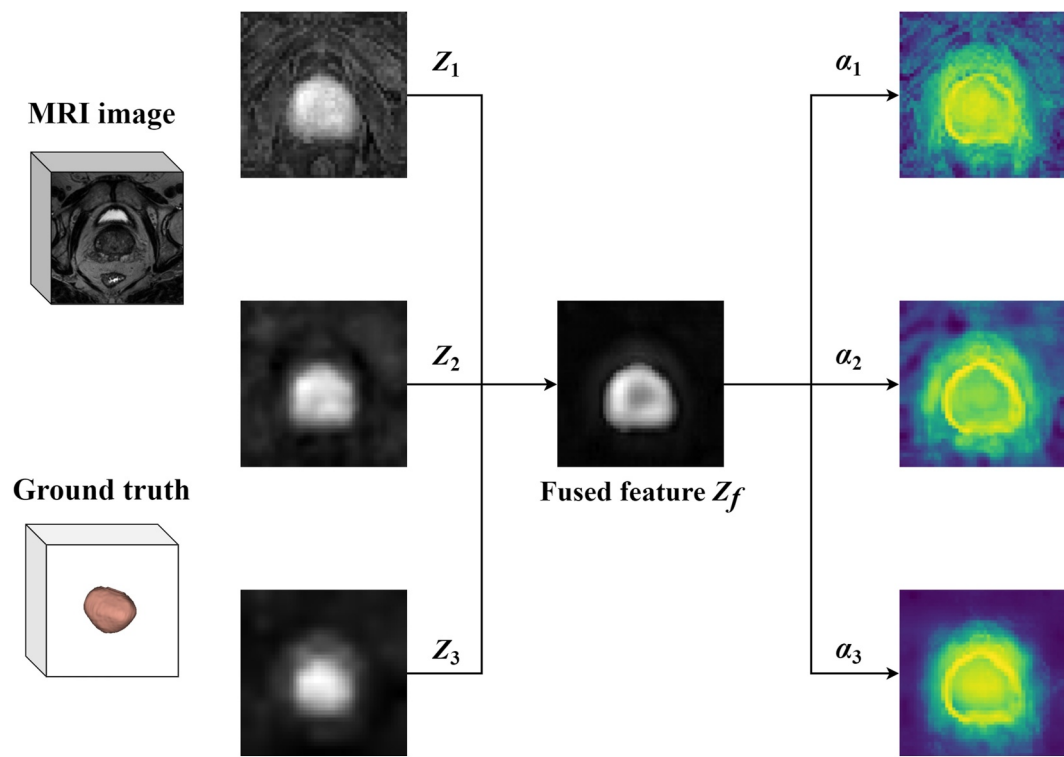


FIGURE 6 Visualisation of feature maps and attention maps for the MLE block. MLE, multi-scale local enhanced.

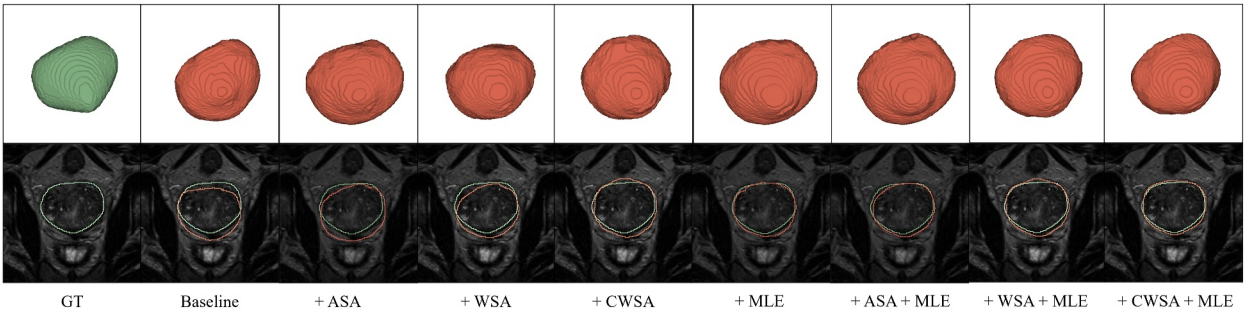


FIGURE 7 Segmented results of the baseline model and its variants on the ProstateX MRI dataset.

the global information effectively and the MLE module can enhance the local information. Secondly, the information interaction among different branches can facilitate improving segmentation performance. By leveraging the features produced by the last three convolution layers of the encoder, we design three parallel branches to reinforce feature learning. Additionally, in the MLE, we introduce voxel attention mechanism and segmentation prototypes to achieve information fusion across the three branches.

Although our method has demonstrated good segmentation performance in three different organs and two modalities, certain limitations need to be addressed. The CWST module reduces the computational load of the model, but it does not reduce the need for GPU memory. Moreover, our method has only focused on the segmentation tasks of single organ (i.e. prostate, heart and pancreas), and does not address multi-organ segmentation and tumour segmentation. Additionally, the direct $4 \times$ upsampling performed in our segmentation head may lead to the loss of image details when dealing with small organs or discrete tissues. In future work, we plan to explore approaches to reduce GPU memory for the SPCT model and extend it to multi-organ and tumour segmentation tasks.

5 | CONCLUSION

This work has presented the SPCT model, a novel serial and parallel network that effectively combines CNNs and transformers for volumetric segmentation tasks. Our method combines strengths of both convolution and self-attention to enhance feature representation and contextual understanding. The CWST module can effectively capture long-range spatial dependencies without introducing excessive computational complexity. Furthermore, the MLE module promotes cross-scale feature interaction, where the sparse global information produced from the CWST complements the dense local information produced from the MLE. Ablation study confirms the impact of the CWST and MLE modules in segmentation. Experiments on three diverse medical datasets validate the superiority of the SPCT model to the compared six segmentation models. Additionally, our method exhibits lower computational complexity compared with other CNN and transformer hybrid models, which highlights its potential for real-time applications.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All datasets can be accessed online.

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